

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool : Short Answer Text(High)

Assessment Tool Weightage: 30%

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

QUESTIONS:

1. UPS Product Tracking Information System

UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.

ANSWER:

The UPS Product Tracking Information System stores information about shipped items, retail centers, and transportation events. Every shipped item is uniquely identified and tracked from the retail center to its final destination through various transportation events.

Entities and Attributes

Shipped_Item

- Item_No (PK)
- Weight
- Dimensions
- Insurance_Amount
- Destination
- Final_Delivery_Date

Retail_Center

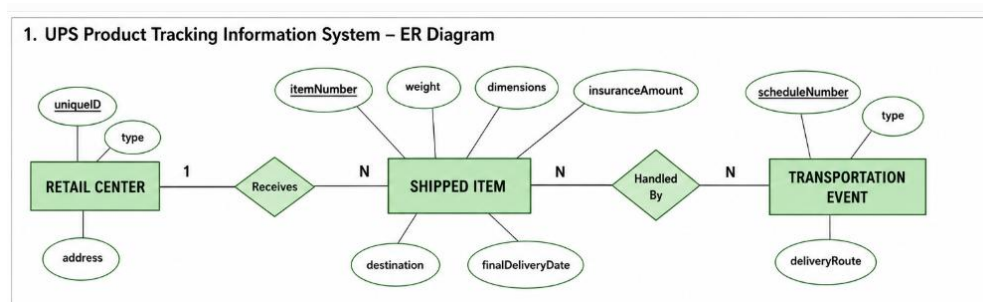
- Center_ID (PK)
- Type
- Address

Transportation_Event

- Schedule_No (PK)
- Type (Flight/Truck)
- Delivery_Route

Relationships

- One Retail Center receives many Shipped Items (1:M).
- One Shipped Item passes through one or more Transportation Events, and one transportation event can carry many shipped items (M:N).



2.Course Management System

Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as

- course with attributes name, department, and c-number
- section with attributes s-number and enrollment, and dependent as a weak entity set on course
- room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.

ANSWER:

A Course Management System is used by a university to schedule classrooms for final examinations. Instead of storing all the information in a single Exam entity, the data is divided into multiple related entities such as Course, Section, Room, and Exam. This reduces data redundancy and improves database organization.

Entities and Attributes

1. Course

Stores the details of each course offered by the university.

Attributes:

- C_Number (Primary Key)
- Course_Name
- Department

2. Section (Weak Entity)

A section belongs to a particular course and cannot exist without it. Therefore, it is modeled as a weak entity.

Attributes:

- S_Number (Partial Key)
- Enrollment

3. Room

Stores information about examination rooms.

Attributes:

- R_Number (Primary Key)
- Capacity
- Building

4. Exam

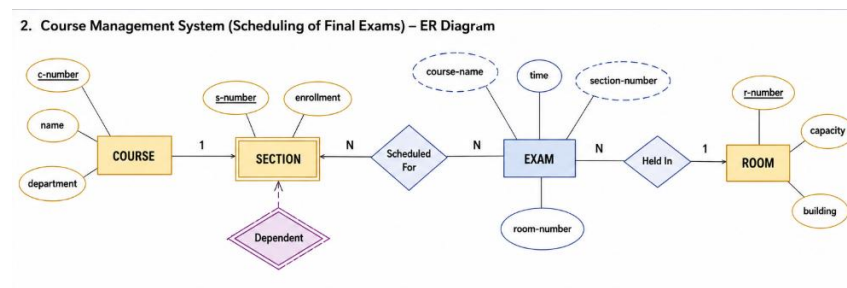
Stores the examination schedule.

Attributes:

- Exam_ID (Primary Key) (*optional*)
- Time

Relationships

- **Course – Has – Section (1 : M)**
 - One course can have many sections.
 - Each section belongs to only one course.
- **Section – Scheduled In – Room (M : 1)**
 - Many sections can be scheduled in the same room at different times.
 - Each section is assigned to one room for the final exam.
- **Section – Has – Exam (1 : 1)**
 - Each section has one final examination.
 - Each examination belongs to one section.



3. University Registrar's Office Management System

A university registrar's office maintains data about the following entities:

- (a) courses, including number, title, credits, syllabus, and prerequisites;
- (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom;
- (c) students, including student-id, name, and program;
- (d) instructors, including identification number, name, department, and title.
- (e) Further, the enrollment of students in courses and grades awarded to students in each

course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.

ANSWER:

A University Registrar's Office Management System maintains information about courses, course offerings, students, instructors, and student enrollments. The registrar's office is responsible for managing course schedules, assigning instructors, enrolling students, and recording grades. The ER diagram organizes these entities and their relationships to ensure efficient storage and retrieval of academic information.

Entities and Attributes

1. Course

Stores the details of each course offered by the university.

Attributes:

- Course_Number (Primary Key)
- Title
- Credits
- Syllabus
- Prerequisites

2. Course_Offering

Represents a particular offering of a course in a specific semester and year.

Attributes:

- Offering_ID (Primary Key)
- Year
- Semester
- Section_Number
- Timings
- Classroom

3. Student

Stores information about students enrolled in the university.

Attributes:

- Student_ID (Primary Key)
- Name
- Program

4. Instructor

Stores information about faculty members who teach courses.

Attributes:

- Instructor_ID (Primary Key)
- Name
- Department
- Title

5. Enrollment (Relationship with Attribute)

Enrollment represents the many-to-many relationship between Student and Course_Offering.

Attribute:

- Grade

Relationships

1. Course – Offers – Course_Offering (1:M)

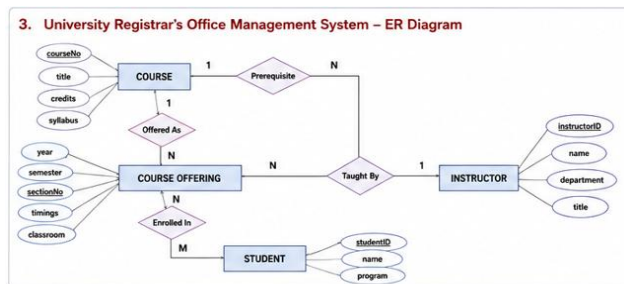
- One course can have many course offerings.
- Each course offering belongs to one course.
-

2. Instructor – Teaches – Course_Offering (1:M)

- One instructor can teach many course offerings.
- Each course offering is taught by one instructor (or multiple instructors if required).

3. Student – Enrolls In – Course_Offering (M:N)

- A student can enroll in many course offerings.
- A course offering can have many students.
- The Enrollment relationship stores the Grade obtained by the student.



4. Hospital Management System

Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)

ANSWER:

A Hospital Management System is used to maintain information about patients, physicians (doctors), medical tests, examinations, and patient history. It helps the hospital store patient records, assign doctors, record medical tests, and maintain a history of treatments and examinations.

Entities and Attributes

1. Patient

Stores the details of patients admitted to the hospital.

Attributes:

- SS# (Primary Key)
- Name
- Insurance

2. Physician

Stores information about medical doctors.

Attributes:

- Physician_Name (Primary Key)
- Specialization

3. Test_Log

Stores details of medical tests and examinations conducted for each patient.

Attributes:

- SS# (Foreign Key)
- Test_Name
- Date
- Time

4. Patient_History

Maintains the previous medical test records of each patient.

Attributes:

- SS# (Foreign Key)
- Test_Name
- Date

5. Doctor_Patient

Represents the relationship between physicians and patients.

Attributes:

- Physician_Name
- SS#

Relationships

1. Physician – Treats – Patient (M:N)

- A physician can treat many patients.
- A patient can be treated by more than one physician.

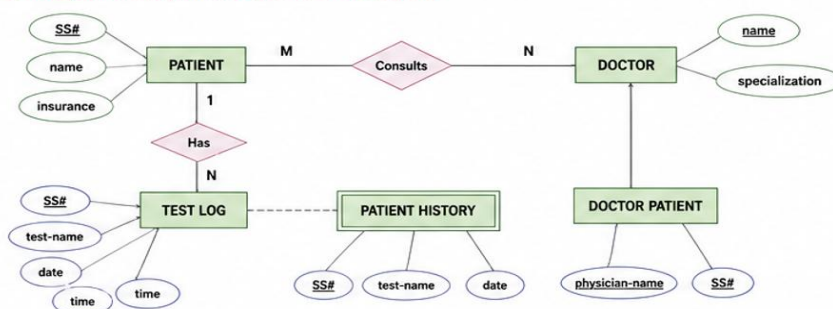
2. Patient – Has – Test_Log (1:M)

- One patient can have many test records.
- Each test record belongs to only one patient.

3. Patient – Has – Patient_History (1:M)

- One patient can have multiple history records.
- Each history record belongs to one patient.

4. Hospital Management System – ER Diagram



5. Company Project Management System

Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations.

□ A department controls a number of PROJECT. Projects have a unique name, number and a single location.

□ Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee.

□ Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).

ANSWER:

A Company Project Management System maintains information about departments, employees, projects, managers, supervisors, and employee dependents. The system helps the company organize departments, assign employees to projects, record working hours, and maintain dependent details for health insurance purposes.

Entities and Attributes

1. Department

Stores information about each department in the company.

Attributes:

- Dept_Name (Primary Key)
- Location

2. Employee

Stores information about company employees.

Attributes:

- SSN (Primary Key)
- Name

- Address
- Salary
- Sex
- Birth_Date

3. Project

Stores details of projects handled by departments.

Attributes:

- Project_No (Primary Key)
- Project_Name (Unique)
- Location

4. Dependent

Stores information about employee dependents for health insurance.

Attributes:

- Dependent_Name
- Birth_Date
- Relationship

Relationships

1. Department – Manages – Employee (1:1)

- Each department is managed by one employee.
- The Start_Date of the manager is recorded.

Relationship Attribute:

- Start_Date

2. Department – Has – Locations (1:M)

- A department may operate at several locations.

3. Department – Controls – Project (1:M)

- One department controls many projects.
- Each project belongs to one department.

4. Department – Employs – Employee (1:M)

- One department has many employees.
- Each employee belongs to only one department.

5. Employee – Works_On – Project (M:N)

- An employee can work on many projects.
- A project can have many employees.

Relationship Attribute:

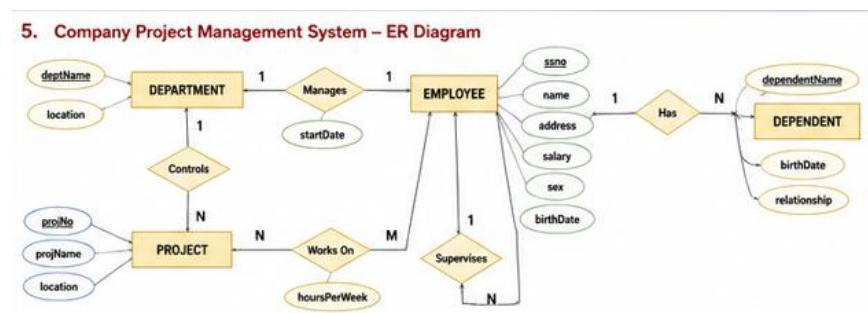
- Hours_Per_Week

6. Employee – Supervises – Employee (1:M)

- One employee (supervisor) supervises many employees.
- Each employee has one immediate supervisor.

7. Employee – Has – Dependent (1:M)

- One employee can have multiple dependents.
- Each dependent belongs to only one employee.



6. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.

ANSWER:

A database consists of two important concepts: Schema and Instance. A schema defines the structure of the database, while an instance represents the actual data stored in the database at a particular point in time.

Schema

Definition

A schema is the overall design or blueprint of a database. It defines the tables, attributes (columns), data types, relationships, and constraints. The schema is created before data is entered and changes very rarely.

Example

Consider a Student table.

Student

RegNo INT (Primary Key)
Name VARCHAR(30)
Department VARCHAR(20)
Age INT

This table structure is the schema of the Student database.

2. Instance

Definition

An instance is the actual data stored in the database at a particular moment. It changes whenever records are inserted, updated, or deleted.

Example :

RegNo	Name	Department	Age
101	Anu	CSE	19

RegNo Name Department Age

102 Rahul ECE 20

103 Priya IT 21

The above records represent one instance of the Student database.

Difference between Schema and Instance

Schema	Instance
Defines the structure of the database.	Contains the actual data stored in the database.
Changes very rarely.	Changes frequently as data is added, updated, or deleted.
Created during database design.	Exists after data is entered into the database.
Also called the database blueprint.	Also called the current state of the database.

Types of Schema**1. Logical Schema****Definition**

The Logical Schema describes the logical structure of the entire database. It defines tables, attributes, relationships, primary keys, and constraints, but does not describe how the data is physically stored.

Example :

Student
RegNo (PK)
Name
Department
Age

Features

- Defines tables and relationships.
- Independent of physical storage.
- Used by database designers and developers.

2. Physical Schema

Definition

The Physical Schema describes how the database is actually stored on storage devices such as hard disks or SSDs. It includes file organization, indexes, storage blocks, and access methods.

Example :

- Student table stored in a database file.
- Index created on RegNo for faster searching.
- Data stored in disk pages.

Features

- Describes physical storage of data.
- Improves database performance.
- Managed by the Database Management System (DBMS).

3. View Schema (External Schema)

Definition

A View Schema defines how different users see the database. It provides only the required data to specific users and hides unnecessary or confidential information.

Example :

A student can view only:

RegNo
Name
Department

The student cannot view salary or confidential staff information.

Features

- Provides customized views for users.
- Improves data security.
- Hides sensitive information.

7. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

ANSWER:

A Bank Management System is used to store and manage information about customers, bank accounts, branches, loans, and transactions. It helps the bank maintain customer records, process deposits and withdrawals, manage loans, and perform secure banking operations.

Entities and Attributes

1. Customer

Stores information about bank customers.

Attributes:

- Customer_ID (Primary Key)
- Name
- Address
- Phone_Number
- Email

2. Account

Stores details of customer bank accounts.

Attributes:

- Account_Number (Primary Key)
- Account_Type
- Balance
- Opening_Date

3. Branch

Stores information about bank branches.

Attributes:

- Branch_ID (Primary Key)
- Branch_Name
- Location

4. Loan

Stores information about loans provided by the bank.

Attributes:

- Loan_ID (Primary Key)
- Loan_Type
- Loan_Amount
- Interest_Rate

5. Transaction

Stores details of banking transactions.

Attributes:

- Transaction_ID (Primary Key)
- Transaction_Date
- Transaction_Type
- Amount

Relationships

1. Customer – Owns – Account (1:M)

- One customer can own one or more accounts.
- Each account belongs to one customer.

2. Branch – Maintains – Account (1:M)

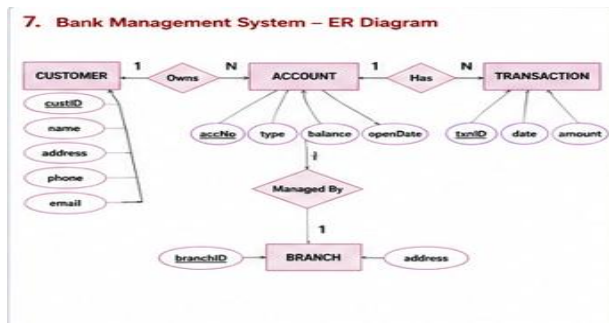
- One branch maintains many accounts.
- Each account is maintained by one branch.

3. Customer – Takes – Loan (1:M)

- One customer can take multiple loans.
- Each loan belongs to one customer.

4. Account – Has – Transaction (1:M)

- One account can have many transactions.
- Each transaction belongs to one account.



8. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.

ANSWER:

The Extended Entity-Relationship (EER) Model is an enhanced version of the Entity-Relationship (ER) Model. It provides additional features to represent more complex real-world situations. Along with entities, attributes, and relationships, the EER model includes concepts such as generalization, specialization, inheritance, and aggregation. These features make database design more flexible, organized, and easier to understand.

1. Extended ER (EER) Model

Definition

The Extended ER (EER) Model extends the traditional ER model by introducing advanced concepts like inheritance, subclasses, and higher-level relationships. It is mainly used for designing large and complex databases.

Features of the EER Model

- Supports inheritance between entities.
- Represents superclass and subclass relationships.
- Reduces data redundancy.
- Improves database organization.
- Models complex real-world applications effectively.

2. Generalization

Definition

Generalization is the process of combining two or more lower-level entities that have common features into a single higher-level entity called a superclass. It is a bottom-up approach, where similar entities are grouped together.

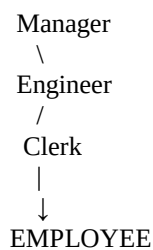
Example :

Suppose a company has the following entities:

- Manager
- Engineer
- Clerk

All of them have common attributes such as Employee_ID, Name, and Salary. These can be combined into a single entity called Employee.

Diagram



Advantages

- Eliminates duplicate data.
- Improves database organization.
- Makes database design simpler.

3. Specialization

Definition

Specialization is the process of dividing one higher-level entity (superclass) into multiple lower-level entities (subclasses) based on their specific characteristics. It is a top-down approach.

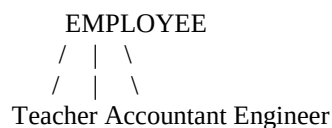
Example :

An Employee entity can be divided into:

- Teacher
- Accountant
- Engineer

Each subclass inherits the common attributes of Employee and also has its own unique attributes.

Diagram



Advantages

- Represents different categories of an entity.
- Supports inheritance.
- Makes data management easier.

4. Aggregation

Definition

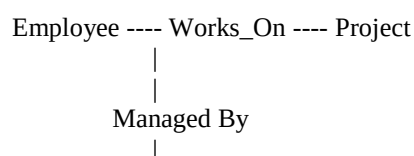
Aggregation is a process in which a relationship between entities is treated as a higher-level entity so that it can participate in another relationship.

It is used when a relationship itself needs to be associated with another entity.

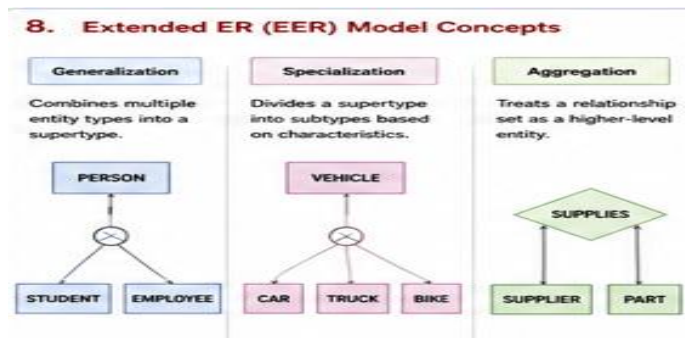
Example

An Employee works on a Project, and this work is managed by a Department. Here, the relationship Works_On is considered as a single entity and connected to the Department.

Diagram



Department



9. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

ANSWER:

A Hospital Management System maintains information about patients, doctors, nurses, appointments, and medical records. The Extended Entity-Relationship (EER) Model is used to represent inheritance (superclass and subclasses) and weak entities.

In this system:

- Person is the superclass.
- Patient, Doctor, and Nurse are subclasses that inherit the common attributes of Person.
- Medical_Record is a weak entity because it cannot exist without a Patient.

Superclass

Person

The Person entity contains common information shared by all people in the hospital.

Attributes:

- Person_ID (Primary Key)
- Name
- Gender
- Age
- Address
- Phone_Number

Subclasses (Inheritance)

1. Patient

Patient inherits all attributes of Person.

Additional Attributes:

- Patient_ID
- Insurance
- Blood_Group

2. Doctor

Doctor also inherits the attributes of Person.

Additional Attributes:

- Doctor_ID
- Specialization
- Qualification

3. Nurse

Nurse inherits the attributes of Person.

Additional Attributes:

- Nurse_ID
- Shift
- Experience

Weak Entity

Medical_Record

Medical_Record depends on the Patient entity for its existence.

Attributes:

- Record_No (Partial Key)
- Diagnosis

- Treatment
- Date

Since a medical record cannot exist without a patient, it is represented as a weak entity.

Relationships

1. Person → Patient, Doctor, Nurse (Inheritance / ISA Relationship)

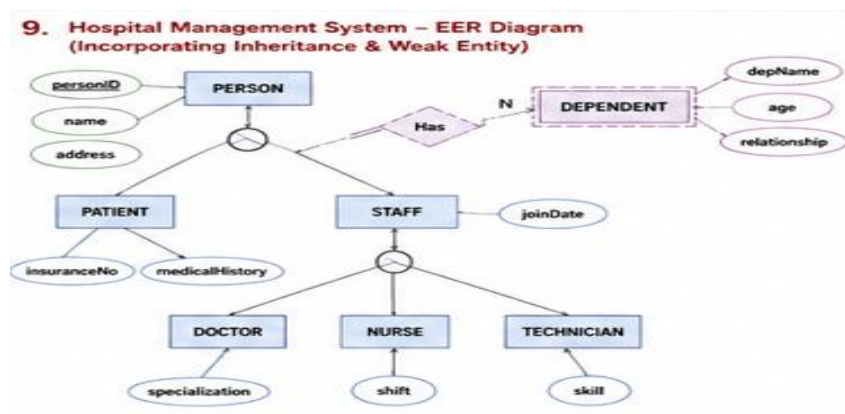
- Patient, Doctor, and Nurse inherit common attributes from Person.

2. Doctor – Treats – Patient (1:M)

- One doctor can treat many patients.
- A patient may be treated by one or more doctors.

3. Patient – Has – Medical_Record (1:M)

- One patient can have many medical records.
- Each medical record belongs to only one patient.



10. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

ANSWER:

A Database Administrator (DBA) is a professional who is responsible for managing, maintaining, and securing an organization's database. The DBA ensures that the database operates efficiently, remains secure, and is always available to users. The DBA also performs

backup and recovery, monitors performance, and protects the database from unauthorized access.

Role of a DBA

The main role of a Database Administrator (DBA) is to ensure that the database is reliable, secure, and performs efficiently. A DBA manages the database throughout its lifecycle, including installation, maintenance, security, backup, recovery, and performance tuning. The DBA also supports developers and end users by ensuring smooth database operations.

Responsibilities of a DBA

1. Database Installation and Configuration

- Installs and configures the Database Management System (DBMS).
- Creates and sets up new databases according to organizational requirements.

2. Database Design and Maintenance

- Creates database structures such as tables, indexes, and relationships.
- Modifies the database whenever changes are required.
- Ensures data consistency and integrity.

3. User Management

- Creates user accounts and assigns roles.
- Grants or revokes user permissions.
- Controls access to sensitive data.

4. Backup and Recovery

- Performs regular database backups.
- Restores lost or corrupted data when necessary.
- Develops disaster recovery plans to prevent data loss.

5. Database Security

- Protects the database from unauthorized access.

- Implements passwords, authentication, and access control.
- Monitors security threats and prevents data breaches.

6. Performance Monitoring and Optimization

- Monitors database performance regularly.
- Optimizes queries and indexes for faster execution.
- Removes unnecessary data and improves storage efficiency.

7. Data Integrity

- Ensures that the data stored in the database is accurate, complete, and consistent.
- Enforces constraints and validation rules.

8. Storage Management

- Monitors disk space and database growth.
- Allocates storage efficiently.
- Maintains database files and indexes.

9. Troubleshooting

- Identifies and resolves database errors.
- Fixes performance issues and system failures.
- Ensures minimum downtime.

10. Support and Documentation

- Assists developers and users in database-related tasks.
- Maintains documentation of database configurations, backups, and security policies.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand